intro_cc_final_project

Submission

1. Requested **screenshot** information below
2. **URL to the Web application.**

Team

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Project Overview

In this project, our team performed analysis of **The Movie Database (TMDB) data.** This project features a medallion-style pipeline (Bronze -> Silver -> Gold) ingests movie metadata and credits, materializes typed Parquet tables, and produces a modeling-ready Gold table with budget, revenue, ROI, release month, genres, and lead director, cast, and production company. Two gradient-boosted regression models predict **box-office revenue** (log-transformed) and **audience rating**, using leakage-aware encoding for high-cardinality categories. Analysis are served on a Streamlit application with data exploration, visual analytics (genre ROI, directors, seasonality, production companies), and interactive prediction.

Technical Stack

- **Datastore** - Datalake on **AWS S3** following medallion architecture
- **Interactive Web page** - Streamlit application hosted on **AWS EC2**
- **Data loading** - Data pull on Streamlit will pull fresh data from Gold folder on S3
- **ML Model Application** - 2 gradient-boosted models for predicting user rating and revenue of a hypothetical movie

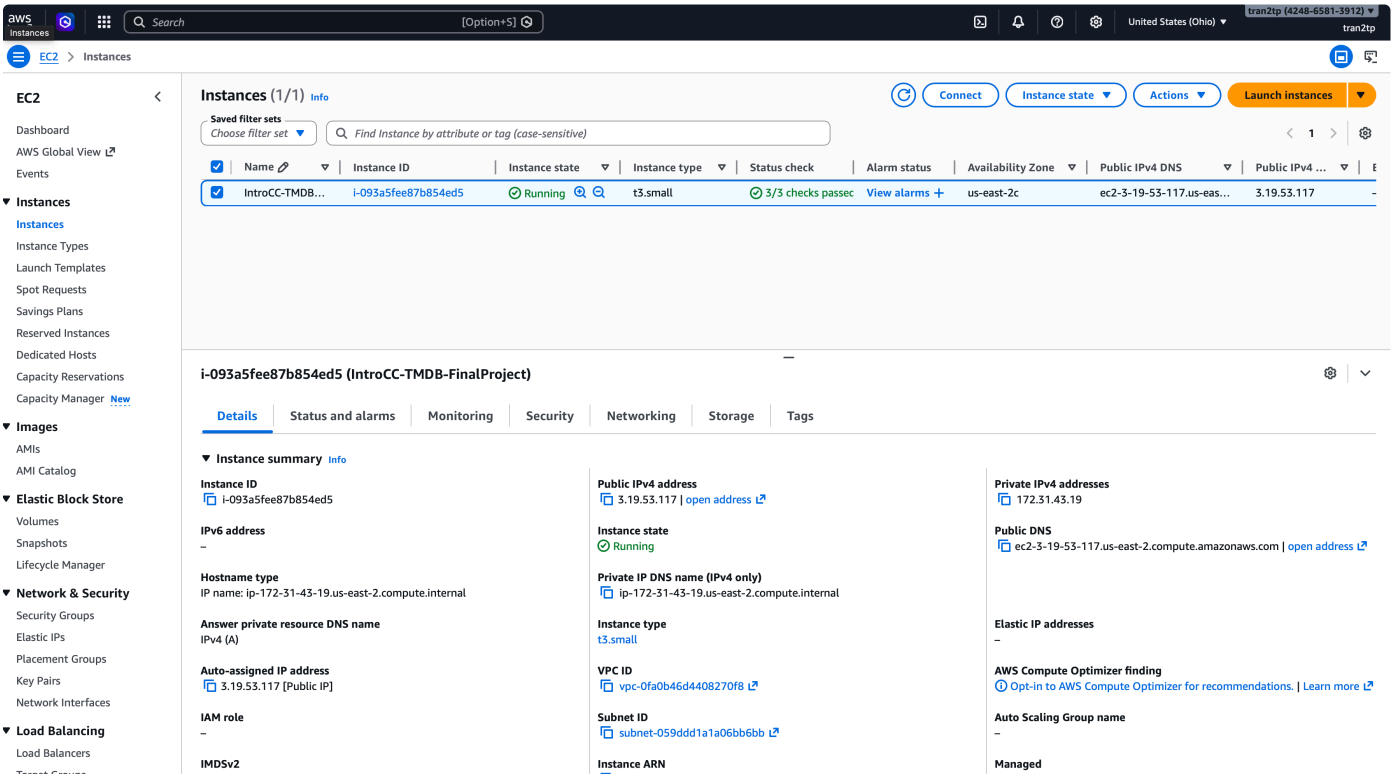
Report

1 Write-Up on ML Models. (Point: 1)

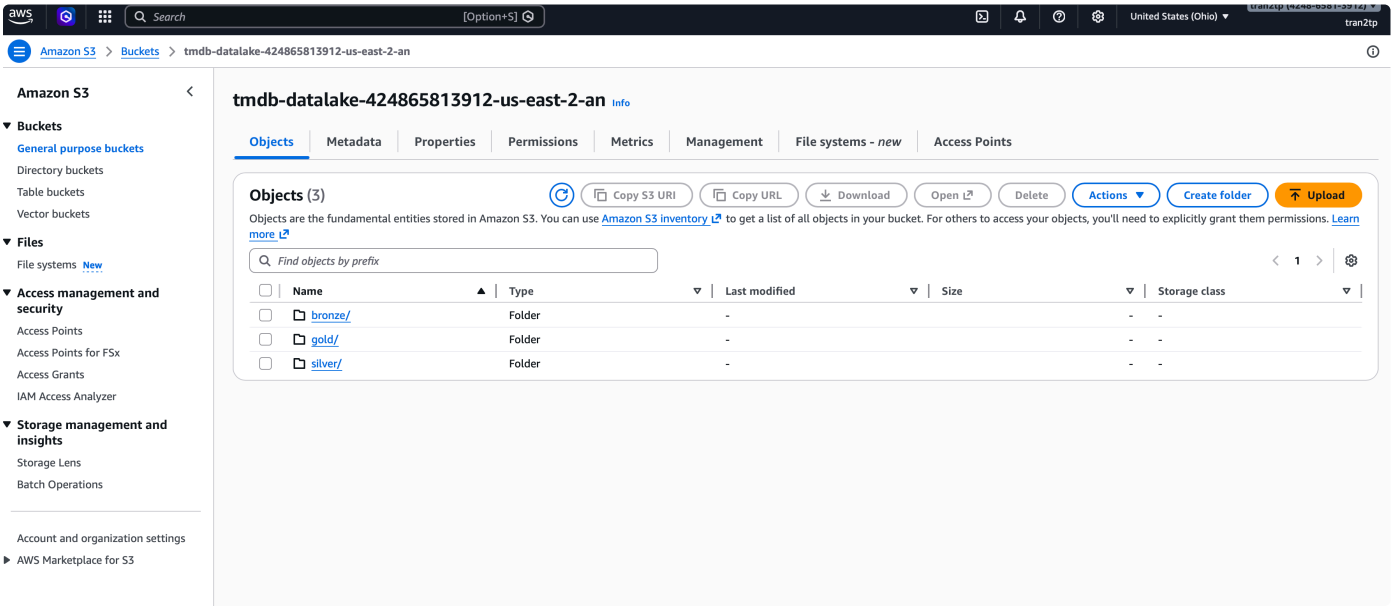
1. **Linear Regression** - This ML model basically tries to draw a best fitting straight line that would resulted in minimal vertical distance to all of the data sample. It works on the assumption that there is a linear relationship between the input features and the target variable. It is highly interpretable, fast to train, and serves as a great baseline, but it may struggle with complex, non-linear relationships.
2. **Random Forest** - This is an ensemble decision tree ML model typically used for classification or regression. It works by aggregating the outputs of all its decision trees (taking a majority vote for classification or an average for regression). By combining multiple models (ensemble), it significantly reduces the risk of overfitting compared to a single decision tree and handles complex, high-dimensional data beautifully.
3. **Gradient Boosting** - This is an ensemble decision tree ML model. Unlike random forest, gradient boosting builds each decision tree sequentially rather than independently. Each new tree focuses entirely on correcting the errors (residuals) made by the previously built trees. This iterative improvement makes it incredibly accurate and a powerhouse for structured tabular data, though it requires careful tuning and is slower to train.

For our project, we will use **gradient boosting**: `HistGradientBoostingRegressor` - a fast gradient boosting regressor in scikit-learn, optimized for large datasets via histogram binning.

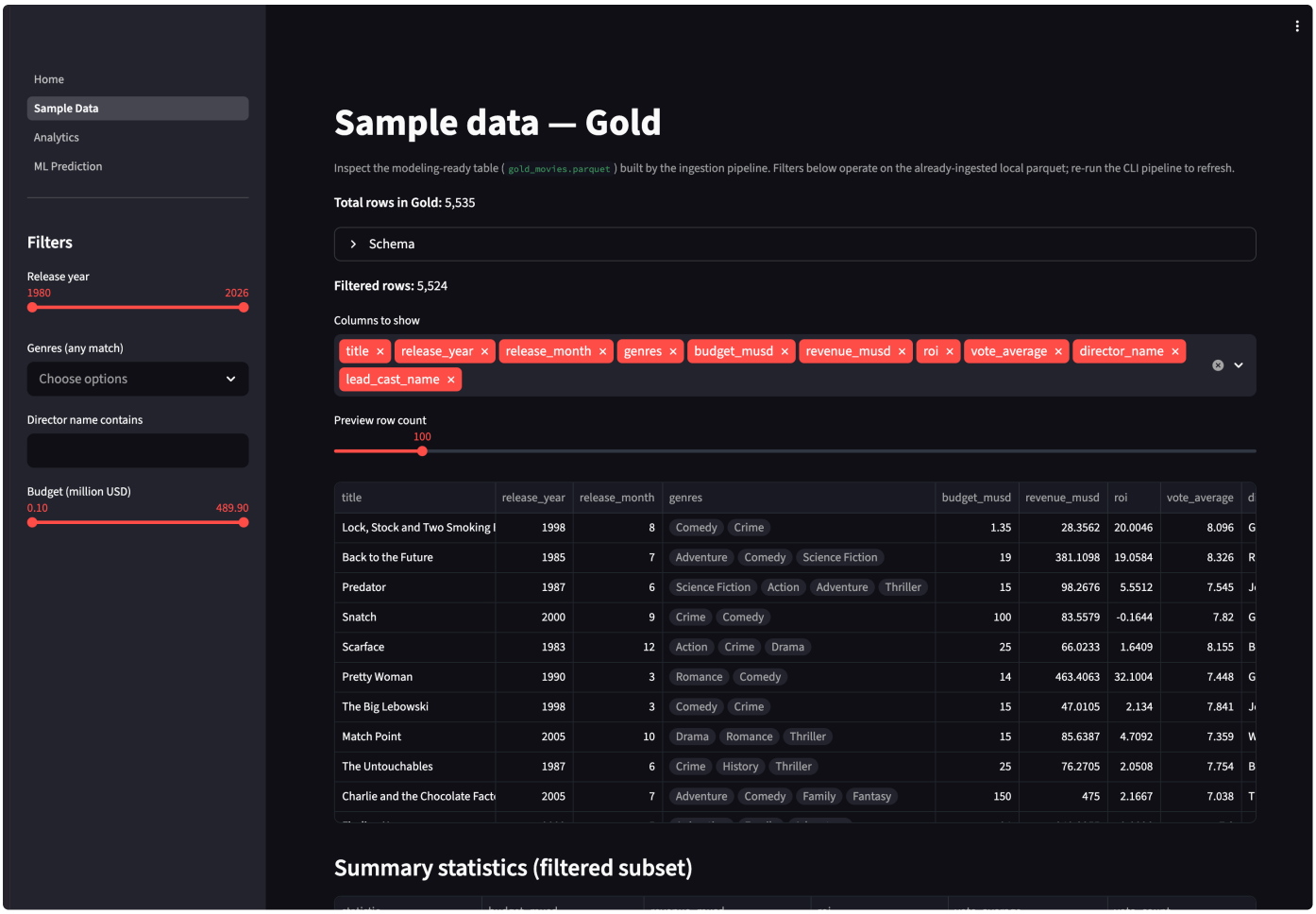
2. Web Server Setup. (Point: 2)



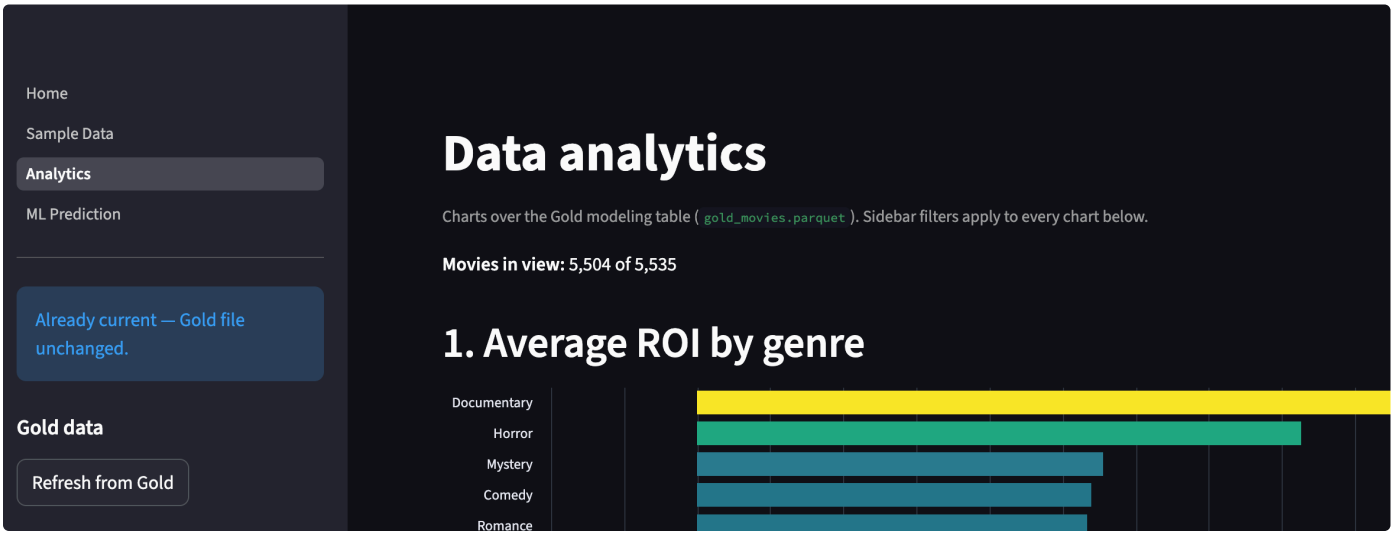
3. Datastore and Data Loading. (Point: 2)



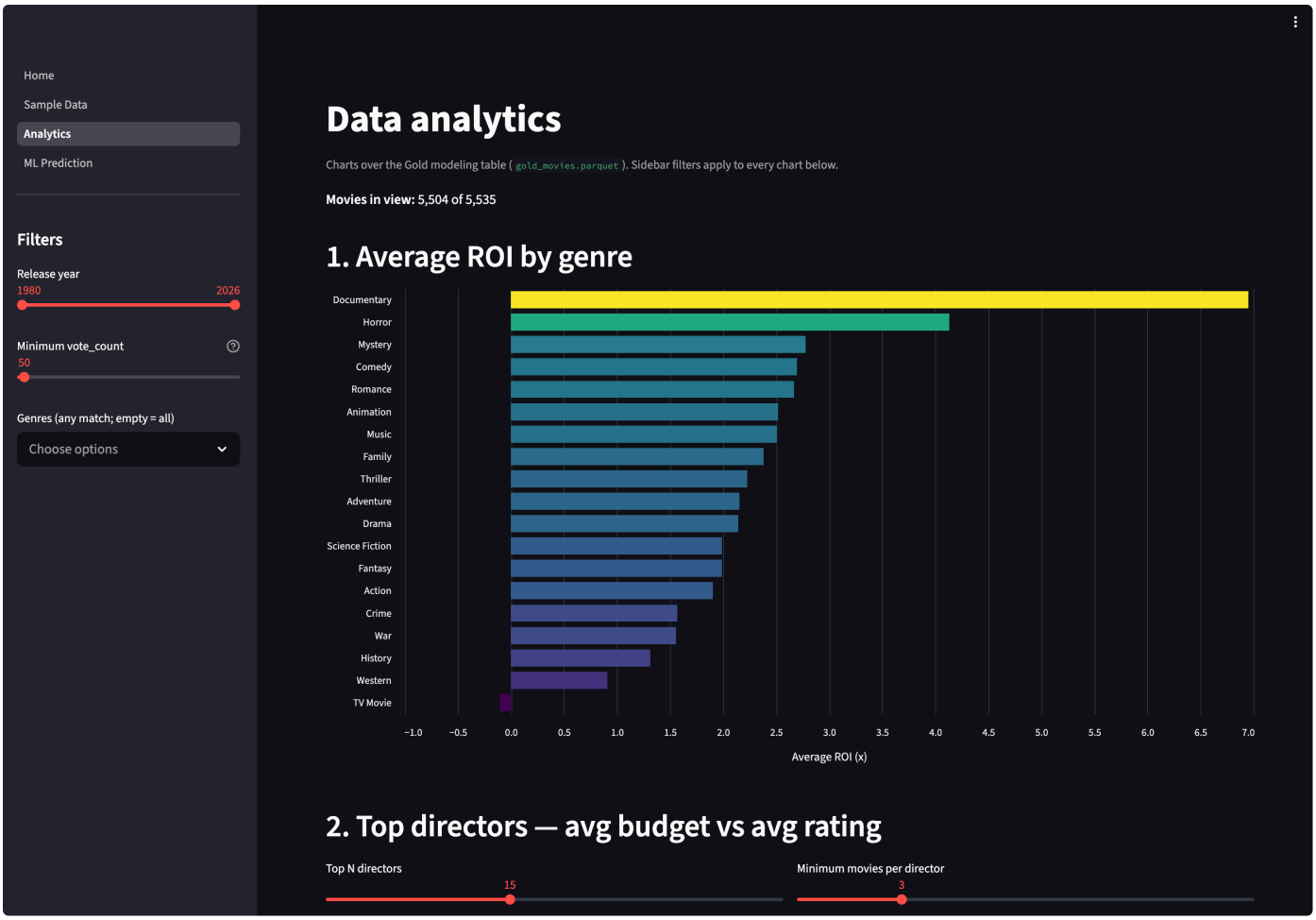
4. Interactive Web Page. (Point: 2)



5. Data Loading Web App. (Point: 2)



6. Web Page with Dashboard. (Point: 2)



7. ML Model Application. (Point: 2)

Home
Sample Data
Analytics
ML Prediction

ML prediction

Two `HistGradientBoostingRegressor` models (revenue & rating) trained on the Gold table with leakage-safe target encoding for director / studio / lead cast. Run `uv run python -m src.ml train` to refresh the artefacts.

Findings | Predict

Revenue model

Target `revenue_musd` (million USD) trained on `log1p(revenue)` so the heavy tail doesn't dominate the loss. Predictions are back-transformed to $M before reporting.

Train rows	Test rows	Holdout R^2
4,428	1,107	0.588

Metrics

Metrics are in the **training space** (log1p for revenue, native for rating). Revenue has an additional $M panel below.

Split	R^2	MAE	RMSE
Holdout (HGB)	0.588	0.856	1.080
5-fold CV (HGB)	0.559	0.848	1.092
Holdout (Ridge baseline)	0.488	0.968	1.203

Revenue back-transformed to $M:

MAE ($M)	RMSE ($M)	MAPE (revenue ≥ $1M)
69.9	141.7	1.23

Permutation importance (holdout, scoring=R^2)

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Findings | Predict

Fill in the hypothetical movie's attributes. Unseen director / studio / cast names fall back to the global prior learned by the target encoder.

Budget (million USD): 60.00

Director: Steven Spielberg

Runtime (minutes): 110

Lead production company: Paramount Pictures

Release month: Jun

Lead cast: Tom Hanks

Genres: Action × Adventure ×

Predict

Predicted revenue	Predicted rating	Implied ROI
$188.4M	6.82 / 10	2.14x

Revenue uses the HGB model trained on `log1p(revenue)` and is back-transformed with `expm1`. Rating is clipped to [0, 10]. ROI is computed from the predicted revenue and the budget you entered.